

make

Lecture 17

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make

- build tool for the creation of software / libraries / documents by specifying dependencies
 - Almost any process that has files as input and outputs can be automated via make
- Originally created by Stuart Feldman in 1976 at Bell Labs
- Almost universally available (all flavors of unix / linux / macOS / Windows)
- Dependencies are specified using a text-based `Makefile` with a simple syntax

Makefile

A **Makefile** provides a list of target files along with their dependencies, and the steps necessary to generate each of the targets from the dependencies.

```
1 target1: depend1 depend2 depend3 ...
2     step1
3     step2
4     step3
5     ...
6
7 depend1: depend4 depend5
8     step1
9     step2
10    ...
```

In the above example **target*** and **depend*** are all just files (given by a relative or absolute path).

Makefile (basic example)

```
1 paper.html: paper.qmd fig1/fig.png fig2/fig.png
2     quarto render paper.qmd
3
4 fig1/fig.png: fig1/fig.R
5     Rscript fig1/fig.R
6
7 fig2/fig.png: fig2/fig.R
8     Rscript fig2/fig.R
```

Smart Building

Because the `Makefile` specifies the dependency structure `make` knows when a file has changed (by examining the file's modification timestamp) and only runs the steps that depend on the file(s) that have changed.

- After running `make` the first time, I edit `paper.qmd`, what steps run if I run `make` again?
- What about editing `fig1/fig.R`?

Variables

Like R or other languages we can define variables

```
1 R_OPTS=--no-save --no-restore --no-site-file --no-init-file .
2
3 fig1/fig.png: fig1/fig.R
4     cd fig1;Rscript $(R_OPTS) fig.R
```

Special Targets

By default if you run `make` without arguments it will attempt to build the *first* target in the `Makefile` (whose name does not start with a `.`). By convention we often include an `all` target which explicitly specifies how to build everything within the project.

`all` is an example of what is called a phony target - because there is no file named `all` in the directory. Other common phony targets:

- `clean` - remove any files created by the Makefile, restores to the original state
- `install` - for software packages, installs the compiled programs / libraries / header files

Optionally, we specify all phony targets by including a line with `.PHONY` as the target and the phony targets as dependencies, i.e.:

```
1 .PHONY: all clean install
```

Builtin / Automatic Variables

- `$@` - the file name of the target
- `$<` - the name of the first dependency
- `$^` - the names of all dependencies
- `$(@D)` - the directory part of the target
- `$(@F)` - the file part of the target
- `$(<D)` - the directory part of the first dependency
- `$(<F)` - the file part of the first dependency

Pattern Rules

Often we want to build several files in the same way, in these cases we can use `%` as a special wildcard character to match both targets and dependencies.

So we can go from

```
1 fig1/fig.png: fig1/fig.R
2     cd fig1;Rscript fig.R
3
4 fig2/fig.png: fig2/fig.R
5     cd fig2;Rscript fig.R
```

to

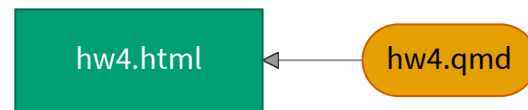
```
1 fig%/fig.png: fig%/fig.R
2     cd $(<D);Rscript $(<F)
```

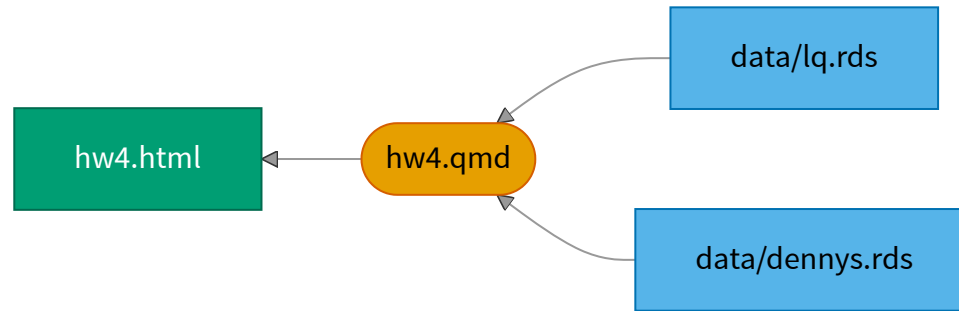
Makefile (fancier example)

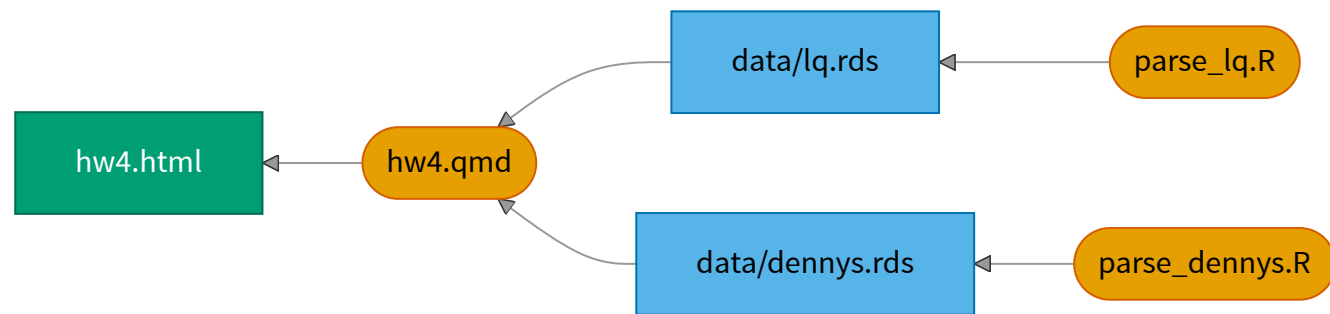
```
1 all: paper.html
2
3 paper.html: paper.qmd fig1/fig.png fig2/fig.png
4     quarto render paper.qmd
5
6 fig%/fig.png: fig%/fig.R
7     cd $(<D);Rscript $(<F)
8
9 clean:
10     rm -f paper.html
11     rm -f fig*/*.png
12
13 .PHONY: all clean
```

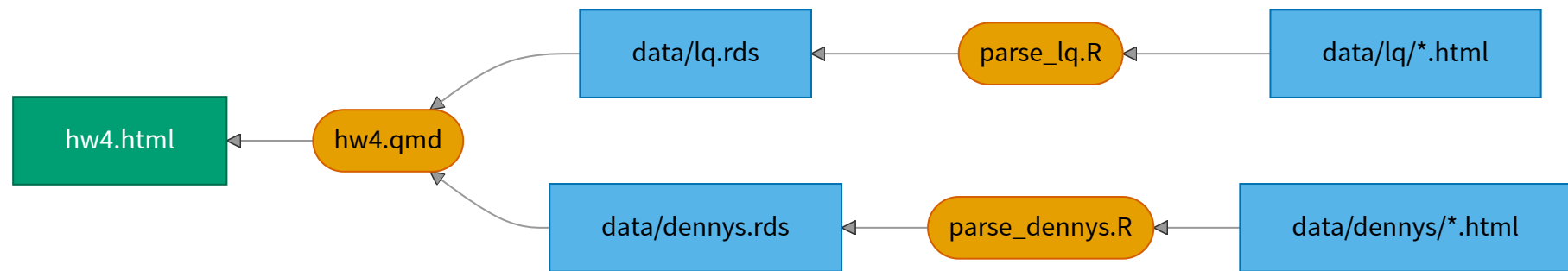
HW4 Makefile

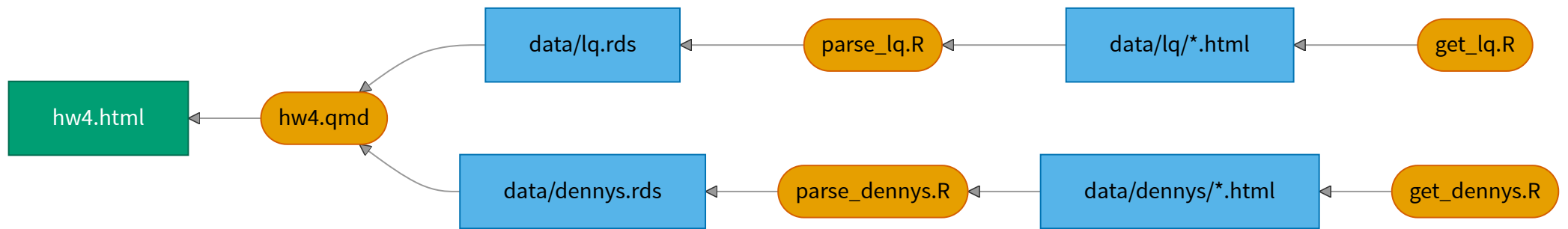
hw4.html











Live Demo

HW4 Makefile

```
1 all: hw4.html
2
3 hw4.html: hw4.qmd data/lq.rds data/dennys.rds
4     quarto render hw4.qmd
5
6 data/lq.rds: parse_lq.R data/lq/*.html
7     Rscript parse_lq.R
8
9 data/dennys.rds: parse_dennys.R data/dennys/*.html
10    Rscript parse_dennys.R
11
12 data/lq/*.html: get_lq.R
13    Rscript get_lq.R
14
15 data/dennys/*.html: get_dennys.R
16    Rscript get_dennys.R
17
18 clean:
19     rm -f hw4.html
20     rm -rf data/
```